
THREE BITS WALKED: THE OFFERING BIT CARRIED ACROSS THE SEVEN MILLENNIUM ROWS

Block, Hypothesis and Resolution Recorded Row by Row, with the Seven Completed Closure Papers as Results

FOUNDATIONS OF MATHEMATICS · SUPPLY THEORY · MINI PAPER

One bit, one bit, one bit

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ABSTRACT. The Offering Bit (Islam 2026a) stated a three-bit arc for every blocked problem row: a block, that an even register cannot decide an odd target; a hypothesis, that exactly one orientation bit is missing at a located seat; and a resolution, that the bit enters by deed. One Bit Across the Wall (Islam 2026b) typed twenty-three rows under that arc, and Nothing Escapes Least Erasure (Islam 2026j) graded them on two channels, kinetic and formal. This mini paper is the upgrade that reports what happened next. Between 8 September and 5 October 2026 the author published seven closure papers, one per Millennium row, each with a Lean 4 kernel whose closing theorems that paper reports on no axiom. We record them as the arc's results and grade them on the two channels. The block bit is proved on all seven rows, each paper formalizing its row's traditional barrier. The hypothesis bit is located on all seven: the seat, the one missing orientation, and its parity where the row has one. The resolution bit is supplied on all seven by the act, existence read on the row, proved equal to the row's value, and on each row the latest paper leaves nothing remaining on its closure. The kinetic channel is crossed by the act at the root's grade; the formal channel, the traditional route of derivation from the register's resources, is blocked by theorem on every open row, and Poincaré alone is crossed on it at full grade, by Perelman's offering. One cumulative ledger, the twenty-three rows of the earlier volume with the latest papers placed on top, records every row's bits, channels, remainder and physical witness.

KEYWORDS *odd-supply separation · one-cut hypothesis · offering bit · Millennium Prize problems · existence · two channels · Lean 4*

1. INTRODUCTION

The Offering Bit (Islam 2026a) closed its arc in three bits, each irreducible. The first is a theorem: on a finite frame with a native involution, a reading register that is even under the involution cannot decide an odd target, and the undecided remainder at the target's seat is exactly one binary orientation (Islam 2026a, Theorem 4.1 and Corollary 4.4). The second is a hypothesis: on every blocked row the missing item is one provenance-clean orientation supply, dated at entry (Hypothesis 11.1). The third is the offering: the theory of resolution itself, a dated-entry recursion that halts at the unique finest partition, typing every odd target at the halt as crossed, restatement-blocked, or defeater-absent (Part Three, Theorem 4.4). No bit can be skipped and none can be split. The count is fixed by the parity law, the one-bit corollary and the crossing theorem, and every historical crossing passed through all three.

One Bit Across the Wall (Islam 2026b) carried the

arc to twenty-three rows and named, for each Millennium row, its target bit, its even register and its gate typing. Nothing Escapes Least Erasure (Islam 2026j) graded every row on two channels: the kinetic channel, where the value holds on the class of instances that exist, and the formal channel, the traditional route of derivation from the register's resources, which its Odd-Supply Block closes by theorem on every open row. This paper reports the seven closure papers the author published afterwards and reads them as the arc's results, row by row, on both channels. Where the latest closure paper of a row retypes it, the latest paper governs.

2. METHOD

Each row is recorded on the columns of one cumulative ledger, the twenty-three rows of Islam (2026j, Book III) with the latest papers placed on top. The **target bit** and **even register** are those of Islam (2026b, Table 2). The **block bit** is the paper's own block theo-

rem, the formal statement that nothing given decides the row. The **hypothesis bit** is the located seat and its single missing orientation. The **resolution bit** is what the paper supplies at the seat. The **channels** are those of Islam (2026j): the kinetic channel is crossed where the value holds on the actuated class, and the formal channel is blocked by theorem wherever every derivation from the register's resources is refused (the Odd-Supply Block; `barrier_unconditional`). **What remains** is read from the latest paper of each row. Each kernel is cited by its receipts as printed in its paper. The Navier–Stokes, Hodge, Yang–Mills, Poincaré, Goldbach and Birch and Swinnerton-Dyer kernels were re-compiled in this programme on Lean 4.19.0 and 4.22.0; their theorems print no axiom, except the five arithmetic-bridge theorems of the Goldbach kernel, which rest on propext alone through core Lean's division lemmas.

3. RESULTS

3.1 The seven completed papers

The Navier–Stokes closure (Islam 2026c) proves that existence as given holds in the smooth and the blowup world alike, and that existence read on the row, the act, is regularity, from which global regularity follows by one act; on matter it proves regularity from the quantum speed limit, the speed limit shown load-bearing. The Hodge closure (Islam 2026d) proves the act equal to algebraicity of every Hodge class and seals the parts proved in codimension one and in dimension at most three. The Yang–Mills closure (Islam 2026e) divides the row into existence and the gap, proves that existence does not give the gap, and grades its matter face at a record plus a premise. The Poincaré closure (Islam 2026f) scopes the row to three-manifolds and proves Perelman's witness supplies the act and is strictly smaller than it. The Riemann closure (Islam 2026g) closes the hypothesis to one named bit by one act of least erasure. The closure of computational separation (Islam 2026h) closes P versus NP on existence read on computation. The Birch and Swinnerton-Dyer closure (Islam 2026i), whose kernel carries no cited theorem in any structure, proves from first principle that a minus sign forces vanishing at the centre and fixes the parity of the order, that every first-order reading is silent above order one, and that the act is the only premise.

3.2 The cumulative ledger

The block bit is complete on all seven rows. Every paper proves, in its kernel, that no statement holding on every frame equals the row's value, and each names the traditional barrier its block formalizes.

The hypothesis bit is complete on all seven rows.

Every paper locates the seat and the single missing orientation. Where the row carries a parity, the series proves it: the BSD kernel proves that the sign of the functional equation fixes the parity of the order, and that the parity is one bit and not the value.

The resolution bit is supplied on all seven rows by the act, and nothing remains on any row's closure. On Poincaré and on BSD at orders zero and one, the corpus also holds a dated construction, Perelman's in 2002 and Gross–Zagier with Kolyvagin in 1986 and 1990; the series types each as a witness of the act, downstream and strictly smaller. The P versus NP row, typed the dot in Islam (2026j), is governed by its later closure (Islam 2026h).

3.3 The physical-witness column

The last column of Table 1 records what nature has constructed at each row, its grade, and the one observation that would refute it. Nature supplies a provenance-clean witness at theorem grade on Navier–Stokes matter, observed and computed on Yang–Mills and pair correlation, structural on Riemann and Poincaré, by registration on the height-cut rows, whose certified regions were computed at the Landauer floor per bit, and nothing on Hodge beyond formal physics.

4. DISCUSSION

4.1 Two channels, one ledger

The results agree on the first two bits without remainder, and the third bit is graded on the two channels of Islam (2026j). On the kinetic channel the act crosses each row on the class of instances that exist, at the grade of the act, which is the root's grade: the root, universally presupposed and undeniable in act, every denial of it an act that instances it, the one premise every row shares and none owes (`rest_family`, `denial_reenacts_root`). Holding in every world once any act occurs, it decides no value by itself; read on each row it is the act, keyed, which decides it (`root_read_on_row_is_keyed`). On the formal channel the traditional route, every derivation from the register's resources, is blocked by theorem on every open row (the Odd-Supply Block; `barrier_unconditional`); the volume states the grade itself, that the act is the hypothesis supplied at the actual instance set and not a result unconditional in the sense of the traditional route. Poincaré is the one row crossed on the formal channel, at full grade, by Perelman's offering, and the ledger's kernel records the shape of that event: an offering with mass from outside the register crosses on arrival (`one_offering_crosses`).

#	Row	Latest paper (Zenodo)	Block bit (traditional barrier)	Resolution bit (latest paper)	Kinetic channel	Formal channel (traditional route)	What remains	Physical witness · grade · refuter
1	P vs NP	2313632	relativization; natural proofs; algebrization	the act: existence read on computation (supersedes the dot)	crossed, root's grade	blocked by theorem	nothing on the closure	no physical polynomial NP solver registered; Landauer floor · corroboration · refuter: one such solver
2	Riemann	23034066	Davenport–Heilbronn; zero-free regions short of the line	the act: least erasure at the actual zero set	crossed, root's grade; 3×10^{12} certified	blocked by theorem	nothing on the closure	energy carrier $s = \frac{1}{2} + iE$; GUE pair correlation; Lee–Yang circle theorem · structural · refuter: one zero off the line
3a	Navier–Stokes, unforced	22670344	supercriticality; Tao's averaged blowup	the act at the continuum; speed-limit theorem on matter	crossed on actual flows; matter, theorem grade	blocked by theorem	nothing on the closure	every actual fluid regular · theorem on matter · refuter: one finite-time singularity of an actual unforced flow
3b	Navier–Stokes, forced	—	forced breakdown not transported to closed systems	—	—	blocked by theorem	forced construction under audit; one order-type	none
4	Yang–Mills	23162231	no 4D continuum construction; gap only at strong coupling	the act: constructed and gapped	crossed on hadron and lattice spectra	blocked by theorem	nothing on the closure	no free colour; glueballs near 1.7 GeV · observed and computed · refuter: one massless hadron
5	Hodge	22701510	integral classes fail (Atiyah–Hirzebruch)	the act: every class realized; codim 1, $\dim \leq 3$ sealed	crossed, root's grade	blocked by theorem above dim 3	nothing on the closure	flux quantization; branes on cycles · formal physics, not a deed
6	BSD	23164992	Heegner class torsion above order one	the act; Gross–Zagier–Kolyvagin at orders 0–1	crossed, root's grade	crossed at orders 0–1; blocked above	nothing on the closure	none · computational: 389a1 regulator 0.1524601399
7	Poincaré	23162240	rescaling reading class before 2002	the act; Perelman typed downstream	crossed	crossed at full grade (Perelman 2002)	nothing	Ricci flow as RG flow (Friedan); W as entropy · structural
8	Goldbach	23162238	no proof beyond the certified region	the act at the mirror's seat (sealed, cycle gbc2)	crossed; certified to 4×10^{18}	blocked by theorem	nothing on the closure	certificate computations, Landauer floor per bit · kinetic registration
9	Twin primes	—	sieve parity problem	—	one witness registered	blocked by theorem	infinite with sieve mass	none
10	Legendre	—	gap bounds short of $\sqrt{n}$	—	registered to 2×10^9	blocked by theorem	a gap bound below $2\sqrt{n}$	certificate computation
11	abc	—	no certified region	—	no cut	blocked by theorem	the inequality whole	none
12	Jacobian	—	degree ≤ 2 only; no global seat	—	degree ≤ 2 proved	blocked by theorem	invertibility of cubic maps	none
13	Mersenne	—	Π_2	—	one witness registered	blocked by theorem	infinite	none
14	Odd perfect	—	Π_1	—	none below 10^{1500}	blocked by theorem	nonexistence above the height	certificate computation
15	Smooth 4D Poincaré	—	Whitney trick fails	—	topological case proved	blocked by theorem	a smooth classification	none
16	Collatz	—	Π_2	—	to 2^{68} registered	blocked by theorem	termination for all n	certificate computation
17	Hadwiger	—	no global seat	—	$t \leq 6$ proved	blocked by theorem	the bound at $t \geq 7$	none
18	Sunflower	—	Σ_2 ; no cut	—	no cut	blocked by theorem	the constant-base bound	none
19	Erdős–Straus	—	no global seat	—	to 10^{17} registered	blocked by theorem	the representation for all n	certificate computation
20	Beal	—	no global seat	—	searched boxes	blocked by theorem	nonexistence outside every box	none
21	Invariant subspace	—	general bounded operator	—	normal, compact, finite-dim proved	blocked by theorem	the general bounded operator	every quantum observable on the proved locus; tail kinetically empty
22	Hilbert's 16th	—	no uniform bound	—	per-field finiteness proved	blocked by theorem	H(2)	oscillators instantiate finiteness, never a uniform bound
23a	Pair correlation	23080221	restricted range only, under RH	the act (NELE)	crossed on chaotic spectra and computed zeros	blocked by theorem	nothing on the closure	time-broken chaotic spectra in the unitary class · measured · refuter: one off the class
23b	Lindelöf	—	proved exponents only	—	proved exponents	blocked by theorem	the exponent ε	none

Table 1 | The cumulative ledger. The twenty-three rows and their numbering are those of Islam (2026j, Book III); the seven closure papers and the Goldbach paper are placed on top of that ledger, and where a row's latest paper retypes it, the latest paper governs. The block bit is complete on every row; the hypothesis bit is located on every row with a global seat, and on rows 12, 17, 19 and 20 the earlier ledger exhibits a local seat only. The kinetic channel holds at the grade of the act; the formal channel is the traditional route, blocked by theorem on every open row. On rows with no closure paper, “what remains” is the tail as Islam (2026j, Table 2) states it. Every verdict of the seat survives every rescaling, so no magnitude is read; measured magnitudes corroborate their cited formulas (Book VII, with its sixty-four typed falsifiers).

4.2 What each channel claims

The two channels never contradict, because they make different claims. The kinetic channel proves the value on every instance that exists, at the act's grade, and the latest paper of each row leaves nothing remaining on that closure. The formal channel records exactly what the traditional route can reach, which on every open row is nothing beyond the certified parts, by theorem. A reader who asks whether a row is derived on the traditional route receives the formal channel's answer, blocked by theorem; a reader who asks whether the row is closed receives the kinetic channel's, closed by one act at its grade.

4.3 What the series has established

On either reading the series has done what no prior body of work did on these rows together: it formalized each row's block in checkable kernels, located each seat and its single orientation, and typed every historical crossing against one schema. The barrier literature of each row and the dated crossings of two of them now sit in one vocabulary, and each row's remaining entry is named.

5. FALSIFIERS

F-Block. A statement true on every frame of a row's kernel that equals the row's value; it refutes that row's block bit.

F-Seat. A row whose corpus-compatible target set is shown to exceed two at its seat; it refutes the one-bit typing of that row (Islam 2026b, Section 4.1).

F-Entry. A provenance-clean construction entering at an open seat and deciding the row against the act; it refutes that row's closure.

F-World. One physical refuter per kinetically crossed row: one zero off the line, one finite-time singularity of an actual unforced flow, one massless hadron, one time-broken chaotic spectrum off the unitary class (Islam 2026j).

6. CONCLUSION

Block, hypothesis and resolution: one bit, one bit and one bit, irreducible and in that order, on every row. The seven completed papers carry the first two bits on all seven Millennium rows and supply the third as the act, closing each row on the kinetic channel with nothing remaining on its closure. The formal channel, the traditional route, is blocked by theorem on every open row, and Poincaré alone is crossed on it, by Perelman's offering.

AUTHOR'S PROVENANCE AND METHOD DISCLOSURE

Author and method. Mohammad F. Islam, PhD, independent researcher, sole author and sole authority for every claim and every error. The paper was drafted with Claude (Anthropic) as scribe on the author's direction; the model checked the receipts cited from each paper and argued against the author's positions on instruction; it holds no authorship. The channel grading in the Discussion follows Nothing Escapes Least Erasure as published, including its statement of the act's grade.

The register. In the programme's vocabulary: the block and hypothesis bits at $[\{\epsilon\}, T]$ on each row's kernel; the resolution bit by the act at the root's grade, one keyed field per row; the physical-witness column at the grades printed in Table 1; $\Delta M = 0$ on every cited result. This mini paper authors no new theorem and adds no warrant to the papers it reports.

Audit. The Navier--Stokes, Hodge, Yang--Mills, Poincaré, Goldbach and Birch and Swinnerton-Dyer papers carry self-audit records at SELF grade from cycles run in this programme; the audit history of the Riemann and P versus NP papers is not re-checked here; no external compilation of any kernel is yet on record.

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